

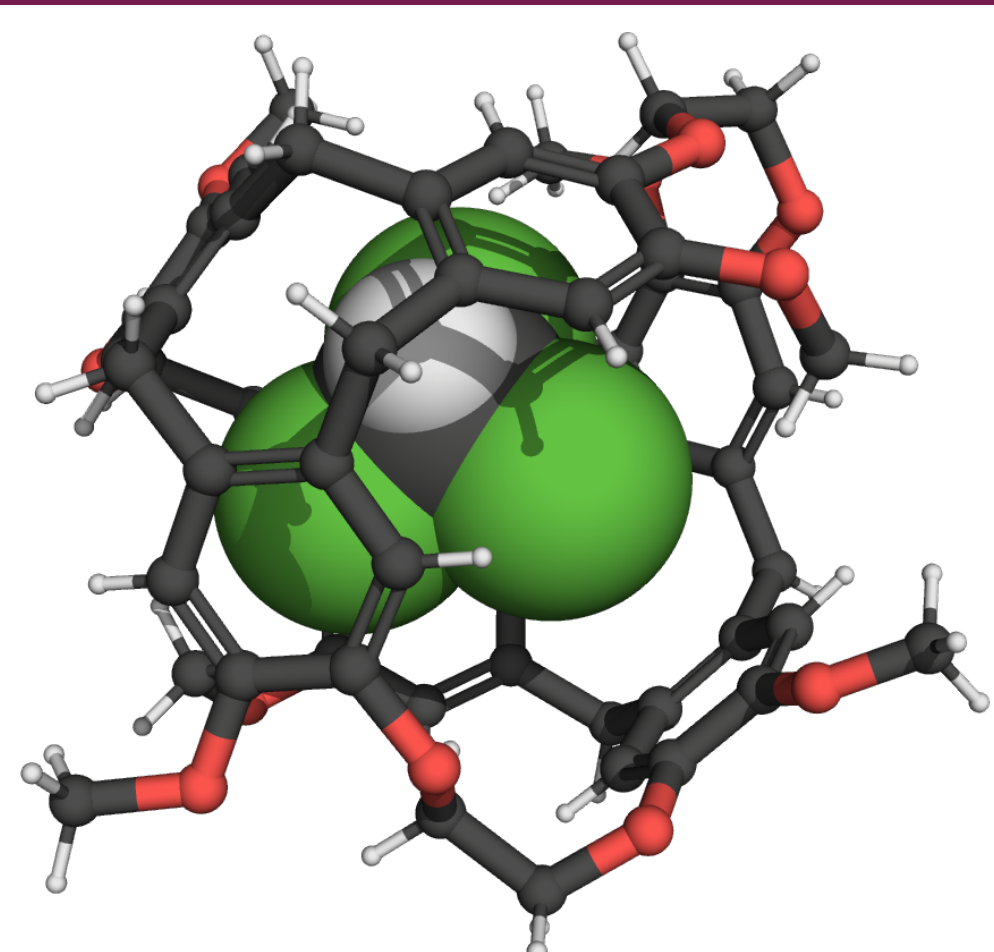


Figure 1. Molecular cage for the chloroform substrate

Molecular cage: Molecule capable of capturing another molecule called a *substrate*.

Criteria:

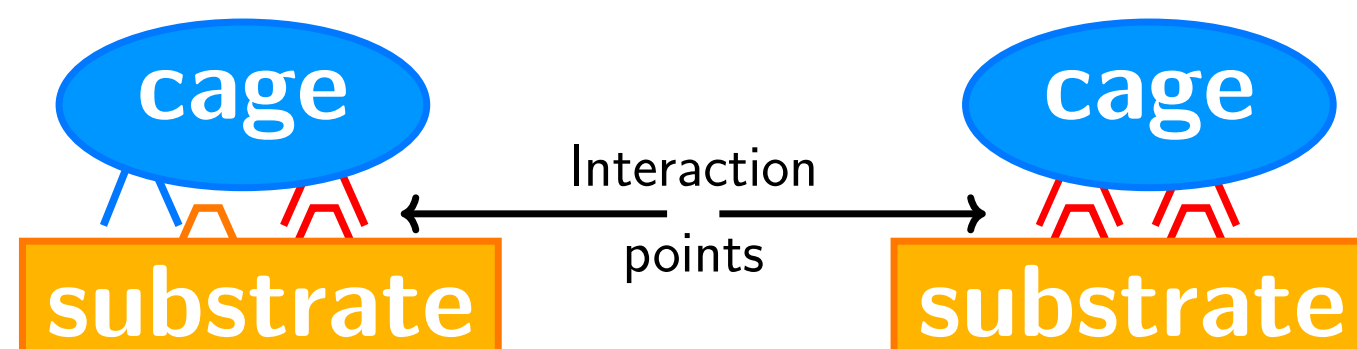


Figure 2. Compatibility: ability to interact with the substrate

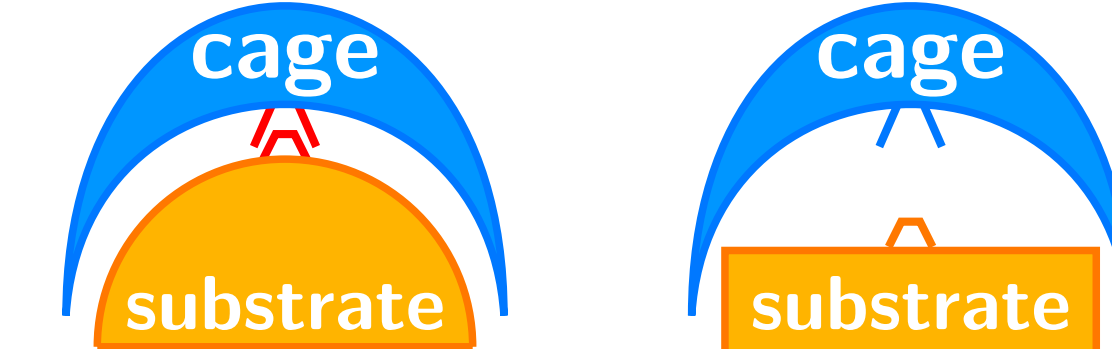


Figure 3. Specificity: interaction with a single substrate

Objective: Generation of a **molecular cage** from a target substrate

Model

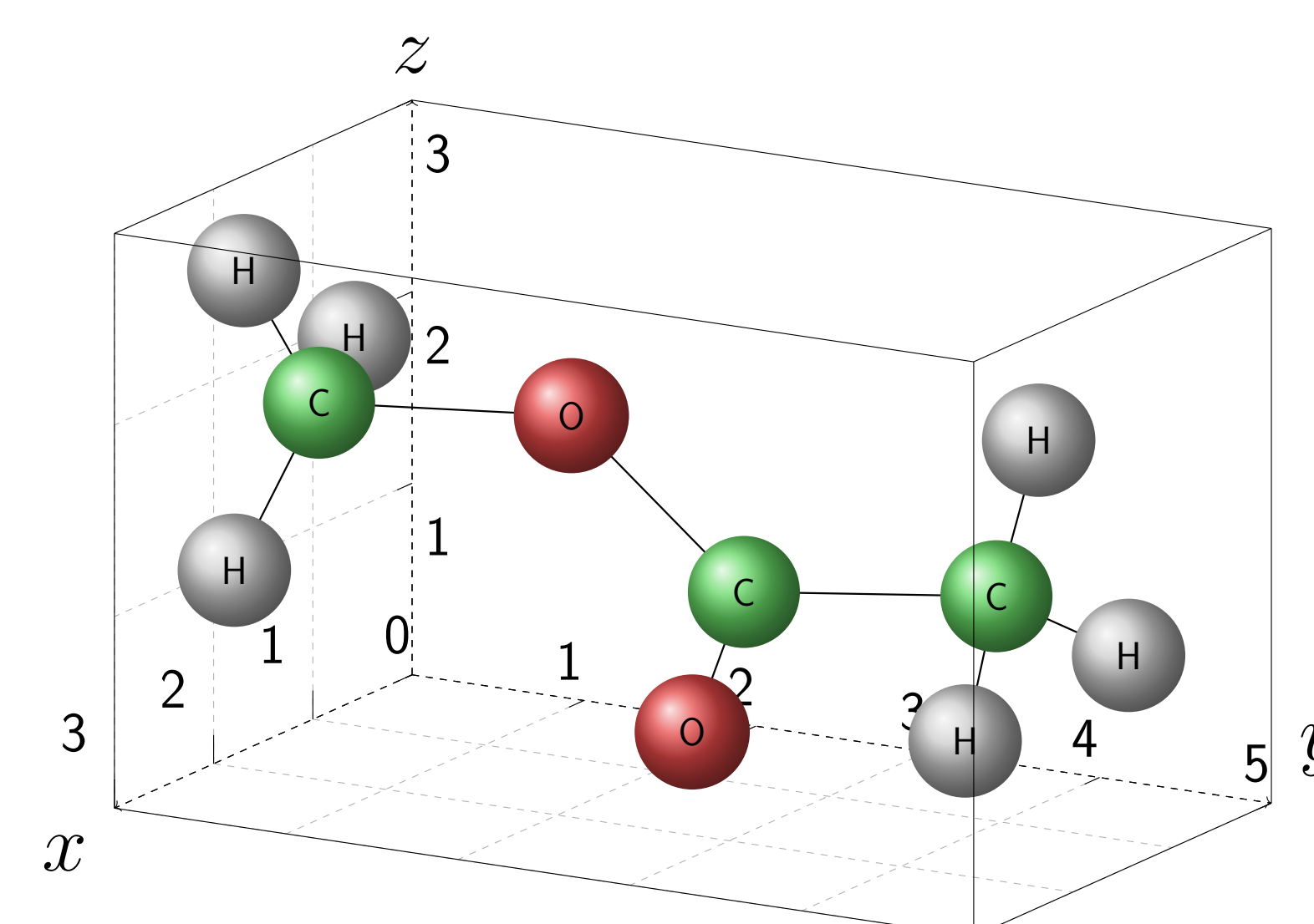
A valid molecular structure is a 3D graph embedding satisfying geometric constraints

Molecular graph: A triple $(G, \text{coord}, \text{type})$, where $G = (V, E)$, $\text{coord} : V \rightarrow \mathbb{R}^3$, $\text{type} : V \rightarrow \text{Atoms}$

Constraints:

- Covalent bond length: $\forall (u, v) \in E, \|u - v\| = d_{\text{covalent}}(\text{type}(u), \text{type}(v))$
- No atomic collision: $\forall u, v \in V \text{ with } (u, v) \notin E, \|u - v\| > d_{\text{collision}}(\text{type}(u), \text{type}(v))$
- Degree constraint: $\forall v \in V, \deg(v) \leq \deg_{\text{max}}(\text{type}(v))$
- Bond angle: $\forall e_1 = (u, v), e_2 = (v, w) \in E, \text{Angle}(e_1, e_2) = \text{Geometry}_{\text{VSEPR}}(\text{type}(v))$

Pattern: Small molecular graph serving as a building block.



Molecular path construction between binding patterns [1]

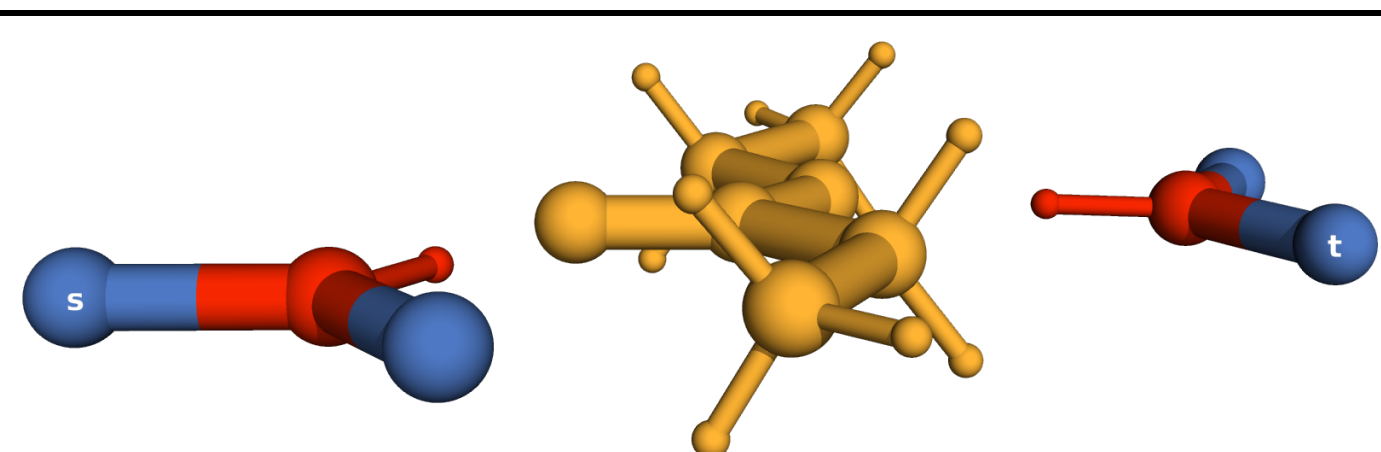


Figure 4. Input [2]: Substrate/orange, Binding patterns/red

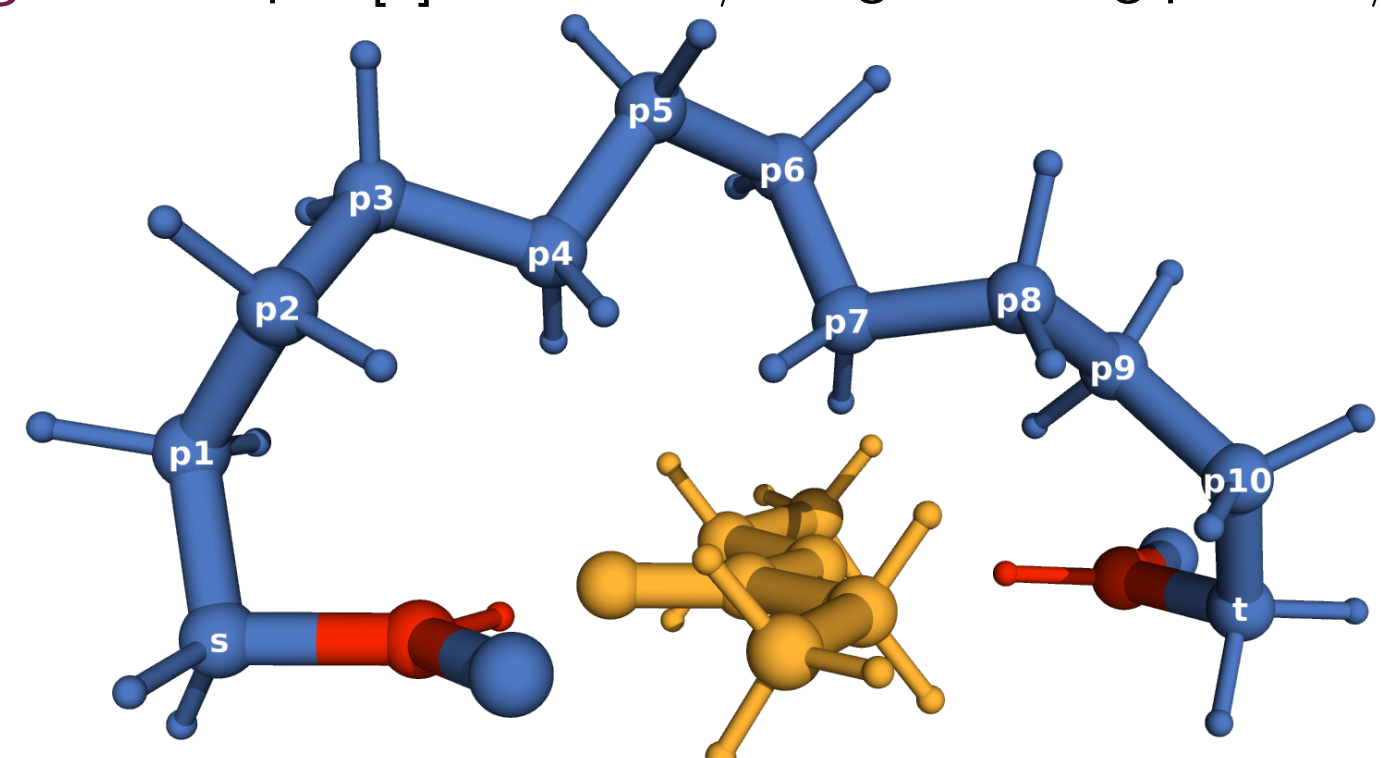


Figure 5. Output: Substrate/orange, Binding patterns/red, Path/blue

Method: Starting from an input (e.g., Fig.4), incrementally connect s to t by placing connection patterns (Fig.6), producing an output (e.g., Fig.5)

Challenges:

- Feasibility constraints**
 - Be a molecular graph
 - Avoid collision
- Search and optimization**
 - Find a feasible path (combinatorial explosion)
 - Minimize the path size / Guide efficiently to t
- Design objectives**
 - Generate diverse solutions
 - Generate cages that are easy to synthesize
 - Generate rigid cages

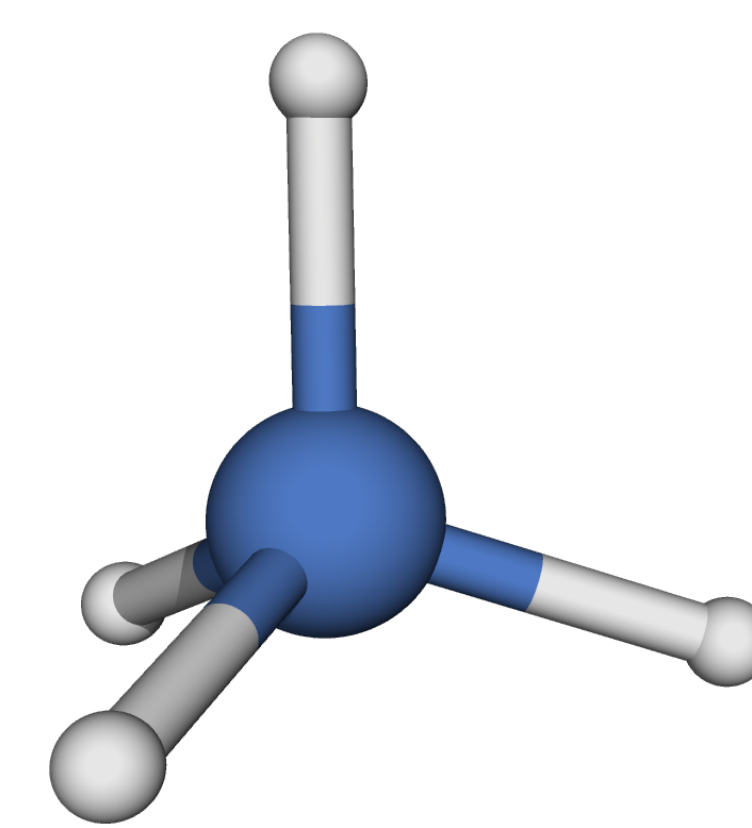


Figure 6. Connection pattern

Finding valid positions for next connection pattern

Bond length + angle constraints $\Rightarrow$ candidates lie on a circle

Valid positions: $\forall u \in V_{\text{substrate}} \cup V_{\text{cage}}$
 $p(\theta) = \text{center} + r \cos(\theta) \vec{v}_1 + r \sin(\theta) \vec{v}_2$
 $\|p(\theta) - u\| > d_{\text{col}}$

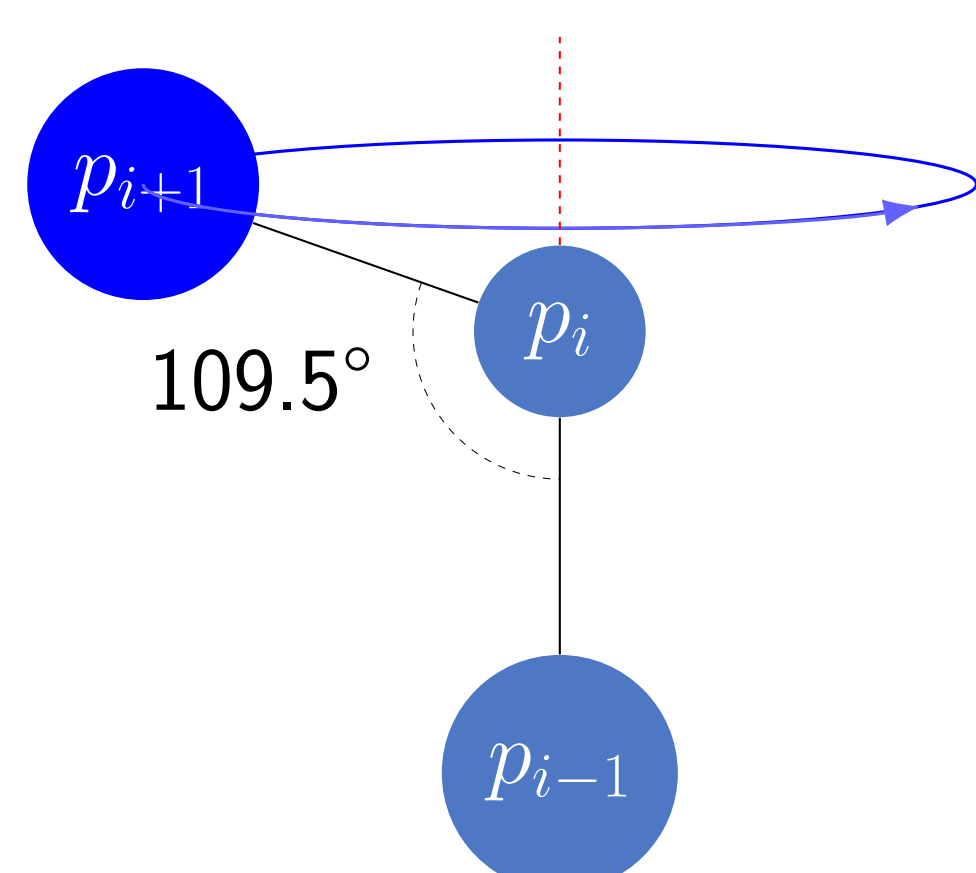


Figure 7. Positions for next connection pattern

From infinite to finite positions:

- Query the grid for atoms near the circle: $\mathcal{O}(1)$ per cell instead of $\mathcal{O}(n)$
- Each atom forbids an angular interval
- Sample positions in the remaining intervals, ranked by distance heuristic

Guide the path toward the target

The path must route around existing atoms

Distance heuristics:

- Euclidean distance (ignores obstacles)
- Grid shortest-path distance (A^*)
- Hybrid: $\begin{cases} \text{Euclidean, target visible} \\ A^*, \text{ otherwise} \end{cases}$

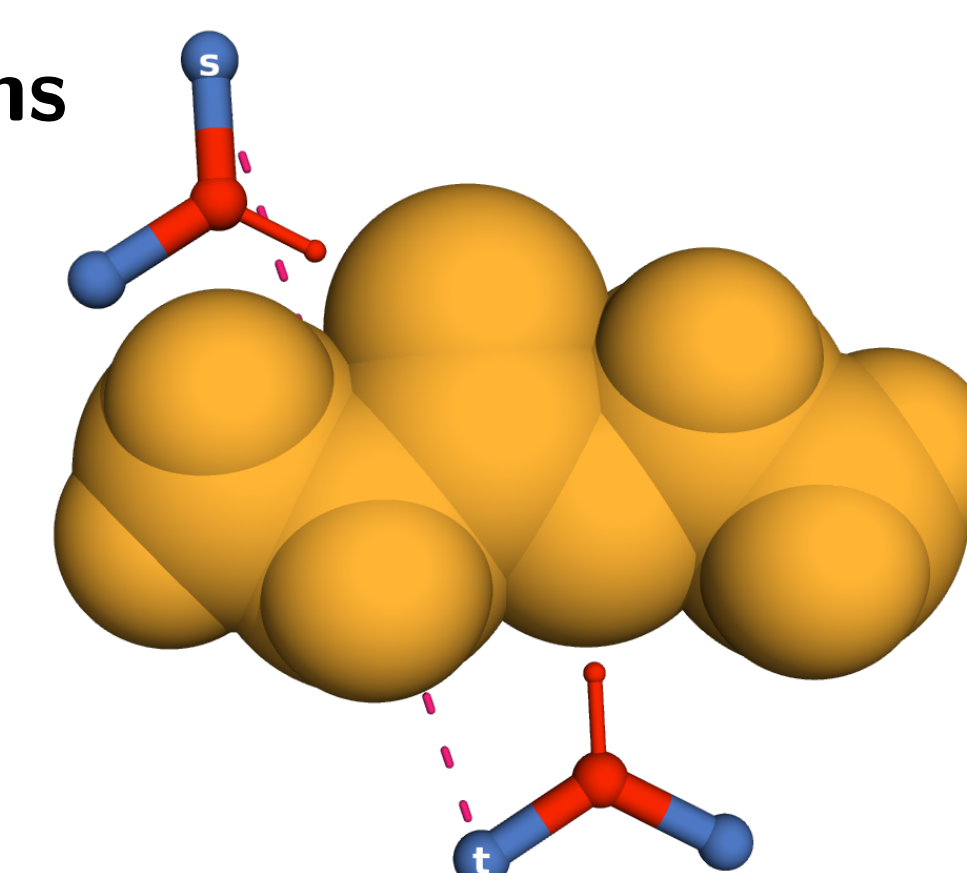


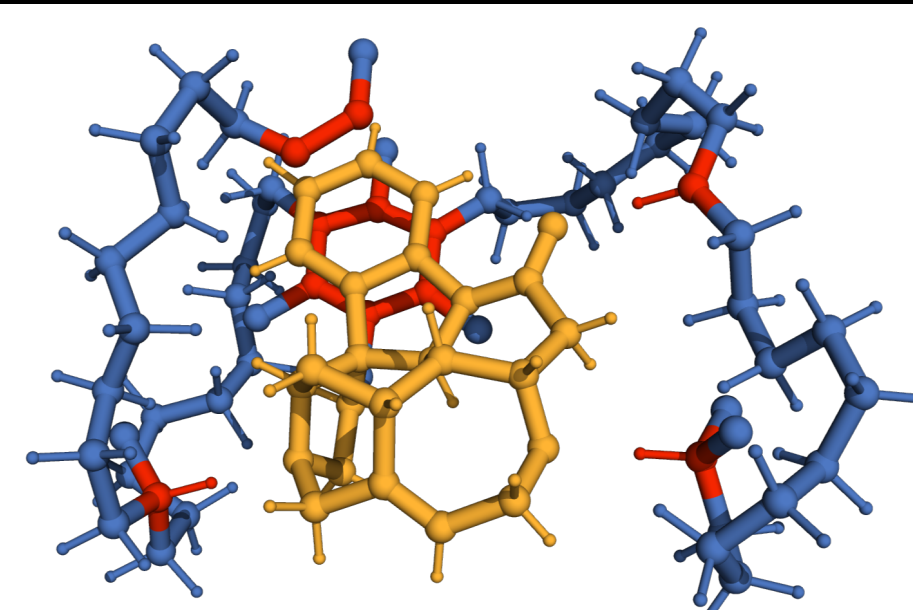
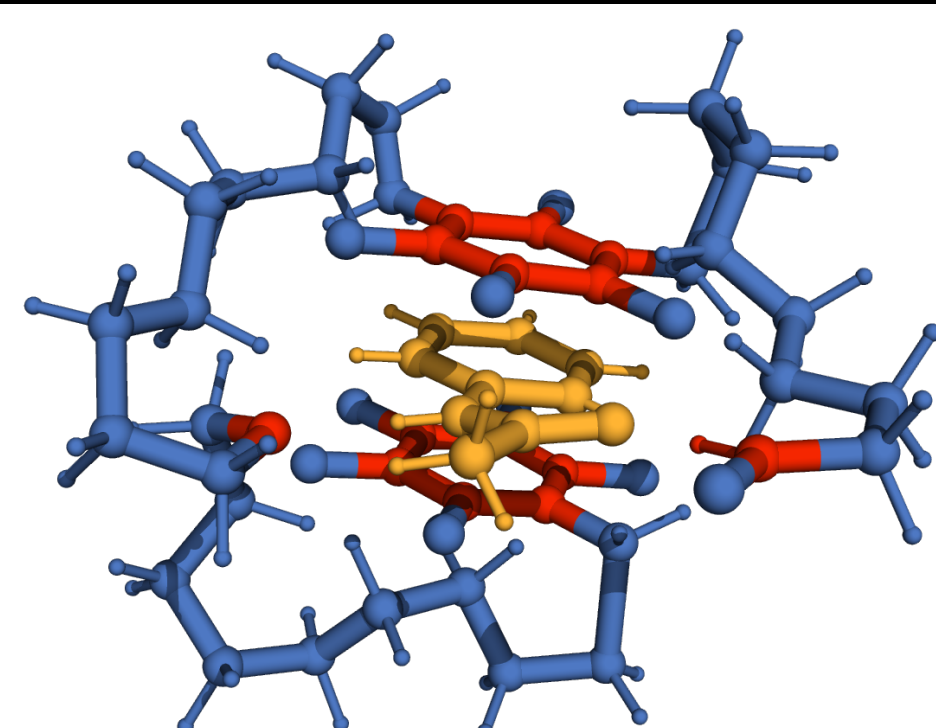
Figure 8. YARZUN03a

Instance	Distance	#Paths	NRMSD	Time (ms)
YARZUN03a	Euclidean	0	NA	0.24
	A^*	29	0.39	78
	HYBRID	26	0.15	29

Exact Euclidean shortest paths avoiding spherical obstacles:
 NP-hard under L_1 ; open under L_2 for disjoint balls

Results

- Satisfy geometric constraints
- Collision-free structures
- Substrate-specific cages



References

Alternative method

Force-directed placement:

Place all atoms at once, then improve their positions with a modified Fruchterman–Reingold algorithm [3]